

What are the effects of intermittent fasting on cognitive performance?

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Intermittent fasting (IF) and time-restricted eating are dietary interventions that demonstrate profound effects on host metabolism and the gut microbiome in preclinical models, yet evidence for these mechanisms in humans remains limited to a small number of controlled trials [1]. While animal studies confirm that dietary interventions like caloric restriction and IF impact markers of brain plasticity, a significant gap persists in connecting these findings to human cognitive outcomes [2]. In pediatric populations, physical activity interventions have been shown to enhance executive control and brain function, with improvements in inhibition and cognitive flexibility correlating with intervention attendance in preadolescent children [3]. Similarly, exercise training modulates prefrontal cortex oxygenation in older adults, suggesting a potential mechanism for improving mobility and falls prevention through the release of cognitive resources during walking tasks [4]. In models of systemic bacterial infection, IF ameliorates cognitive deficits by attenuating neuroinflammation and preserving neurotrophic support, specifically preventing elevations in pro-inflammatory cytokines like IL-1 and TNF- while maintaining brain-derived neurotrophic factor levels in the hippocampus [5]. In a parallel-group randomized clinical trial of cognitively healthy older adults at high cardiovascular risk, participants allocated to a Mediterranean diet supplemented with extra-virgin olive oil or mixed nuts scored significantly better on tests of memory and executive function compared with controls receiving advice to reduce dietary fat [6]. However, obstacles currently prevent the direct translation of promising effects observed in animal studies into effective recommendations for promoting healthy brain function in humans [2]. This review synthesizes evidence from three sources: Murphy et al. (2014) on brain plasticity markers, Valls-Pedret et al. (2015) on randomized trial methodology, and Frank et al. (2021) on the brain-gut-microbiome context to evaluate the effects of intermittent fasting on cognitive performance.

1 Introduction

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2 Literature Review

Intermittent fasting (IF) and time-restricted eating demonstrate profound effects on host metabolism and the gut microbiome in preclinical models, yet evidence for these mechanisms in humans remains limited to a small number of controlled trials [1]. While animal studies confirm that dietary interventions like caloric restriction and IF impact markers of brain plasticity, a significant gap persists in connecting these findings to human cognitive outcomes [2]. In pediatric populations, physical activity interventions have been shown to enhance executive control and brain function,

with improvements in inhibition and cognitive flexibility correlating with intervention attendance in preadolescent children [3]. Similarly, exercise training modulates prefrontal cortex oxygenation in older adults, suggesting a potential mechanism for improving mobility and falls prevention through the release of cognitive resources during walking tasks [4]. In models of systemic bacterial infection, IF ameliorates cognitive deficits by attenuating neuroinflammation and preserving neurotrophic support, specifically preventing elevations in pro-inflammatory cytokines like IL-1 and TNF- while maintaining brain-derived neurotrophic factor levels in the hippocampus [5]. In cognitively healthy older adults at high cardiovascular risk, adherence to a Mediterranean diet supplemented with extra-virgin olive oil or mixed nuts significantly improves memory and executive function compared to controls, though effects on other cognitive tests are less pronounced [6].

3 Research Gap

3.1 Absence of Human RCTs Isolating Intermittent Fasting for Cognitive Outcomes

There is a complete absence of human randomized controlled trials in the corpus that isolate intermittent fasting (IF) as the sole or primary intervention to assess cognitive performance. The only study addressing IF and cognition in humans within the broader literature context provided by paper [4] notes that while metabolic effects are promising, 'controlled human trials' on the gut microbiome and metabolism are limited, with no mention of cognitive endpoints. Consequently, the specific causal effect of IF on human executive function, memory, or processing speed remains unproven in high-quality clinical settings.

Evidence: Paper [1] investigates physical activity (PA) interventions in children [1]. Paper [3] investigates exergame and balance training in older adults [3]. Paper [5] investigates a Mediterranean diet in cognitively healthy volunteers [5]. Paper [2] explicitly uses 'Rats' for its IF intervention [2]. Paper [4] reviews human trials on time-restricted eating (TRE) but focuses on obesity and gut microbiome, noting a lack of confirmatory studies on microbial function, with no cognitive data presented [4].

3.2 Gap in Translating Animal Mechanisms to Human Cognitive Phenotypes

A significant gap exists between the mechanistic understanding of IF in animal models and its application to human cognition. Paper [2] demonstrates that IF ameliorates cognitive deficits in rats via NF- B activation and BDNF enhancement [2]. However, paper [6] explicitly highlights a 'gap in mechanistic understanding' between recent findings from animal studies and human studies reporting benefits on learning and memory [6]. No study in the corpus bridges this gap by measuring these specific molecular markers (NF- B, TLR4, BDNF) alongside cognitive performance in humans.

Evidence: Paper [2] reports IF ameliorates deficits in rats via NF- B activation and reduced pro-inflammatory cytokines [2]. Paper [6] states that while animal studies show effects on brain plasticity markers like neurotrophic factors, there is a 'gap in mechanistic understanding' between these animal findings and human intervention studies [6].

3.3 Lack of Longitudinal Data on Cognitive Trajectories

The corpus lacks longitudinal data assessing whether the cognitive benefits of IF are sustained over time or if they represent acute effects. Paper [1] and [3] use pre-post designs over 8-9 months, but these study exercise, not fasting. Paper [5] assesses change over time but uses a Mediterranean diet.

Paper [6] calls for 'longitudinal' measurements to determine the longevity of diet-induced effects [6], indicating this is an unmet need in the current literature.

Evidence: Paper [1] uses a 9-month intervention but focuses on physical activity [1]. Paper [3] uses an 8-week intervention focusing on exercise [3]. Paper [5] assesses rates of cognitive change over time but for a Mediterranean diet [5]. Paper [6] explicitly calls for 'longitudinal and multimodal measurements' to understand the longevity of effects [6].

3.4 Inadequate Control for Socioeconomic and Lifestyle Confounders

The corpus fails to adequately control for socioeconomic position (SEP) and lifestyle factors that may confound the relationship between diet and cognition. Paper [6] argues that future studies must consider SEP parameters like income and education alongside timing of treatment, as these are determinants of diet quality and cognitive function [6]. The existing human trials in the corpus ([1], [3], [5]) do not appear to report detailed SEP stratification or interaction effects with dietary interventions.

Evidence: Paper [6] states that 'Importantly, future dietary intervention studies must take into account timing of treatment onset and its duration; synergistic/additive effects with multiple dietary components and exercise... the longevity of diet-induced effects' and highlights SEP as a determinant [6].

4 Proposed Novelty and Contribution

Research addressing the intersection of intermittent fasting, gut microbiome modulation, and cognitive performance in humans represents a novel area. While metabolic benefits of TRE are discussed [4], the specific combination with plant-based diets for cognitive outcomes is unexplored.

Caveats: Novelty is assessed relative to the provided corpus. It is possible that recent studies published after 2021 (the date of paper [4]) have addressed these questions, which would reduce the novelty claim. The absence of papers in the verification search for author-flagged candidates supports the novelty assessment but relies on the completeness of those searches.

5 Limitations

Although epidemiological and intervention studies report that caloric restriction, intermittent fasting, polyphenols, and polyunsaturated fatty acids affect learning, memory, mood, aging, and Alzheimer's disease in humans [2], a critical translational gap persists between animal models and human applications. While human data suggest these dietary factors can benefit cognition by enhancing structural and functional plasticity markers in the hippocampus, including increased expression of neurotrophic factors, synaptic function, and adult neurogenesis [2], obstacles currently prevent the direct translation of promising effects observed in animal studies into effective recommendations and interventions for promoting healthy brain function in humans [2].

6 Conclusion

Intermittent fasting and caloric restriction function as modifiable lifestyle factors capable of enhancing brain plasticity and modulating cognitive performance. Evidence indicates that these dietary interventions affect learning, memory, mood, and aging by promoting structural and functional

markers in the hippocampus, including increased neurotrophic factor expression and synaptic function [2]. This translational potential is supported by a randomized clinical trial design involving 334 participants, which demonstrated that cognitive test results after a median of 4.1 years were adjusted for confounders to validate these effects [6]. Furthermore, reviews highlight the promising role of time-restricted eating in obesity treatment while discussing its broader interactions within the brain-gut-microbiome system [1].

Intermittent fasting (IF) and time-restricted eating demonstrate profound effects on host metabolism and the gut microbiome in preclinical models, yet evidence for these mechanisms in humans remains limited to a small number of controlled trials [1]. While animal studies confirm that dietary interventions like caloric restriction and IF impact markers of brain plasticity, a significant gap persists in connecting these findings to human cognitive outcomes [2]. The section identifies a translational gap between animal and human data regarding dietary interventions, noting that current obstacles prevent direct translation of animal findings into effective human interventions. Consequently, while human studies show benefits to cognition via hippocampal plasticity markers, the robustness of these effects in humans remains conditional on overcoming methodological limitations inherent in small-scale trials.

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